# HSR-ACC-27 Access Node Failure Remediation Procedure (Arista)

**Node:** HSR-ACC-27 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_5614

## Description

The failure of Access Node HSR-ACC-27, located at HSR Layout Access, is attributed to hardware/environmental fault within the ASIC memory parity errors, line card crashes, or power supply redundancy loss. This phenomenon occurs due to the intricate relationship between the device's hardware and software components. The Arista Access switch, being a critical juncture in the network fabric, relies on the Line Card (LC) and Power Supply (PSU) to maintain connectivity and provide a stable platform for forwarding packets. In the event of a Line Card crash or Power Supply redundancy loss, the ASIC's memory parity errors may escalate, triggering the ALARM_5614. This alarm signifies a critical warning that necessitates immediate remediation to prevent network instability and potential data loss. A thorough understanding of the underlying hardware and software mechanics is essential to diagnose and resolve this issue effectively.

## Pre-requisite Checks

- Verify Site Location: HSR Layout Access
- Ensure Remote Hands are dispatched to the site
- Check redundant link status and network topology
- Perform an initial assessment of the Line Card and Power Supply health

## Potential Root Causes

| Cause | Probability |
|---|---|
| ASIC memory parity errors | High |
| Line Card crash | Medium |
| Power Supply redundancy loss | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show system internal memory detail, Expected Output: Memory details, including parity errors and Line Card health, Description: This command provides an in-depth view of the system memory, including parity errors, which can help diagnose ASIC-related issues.

Command: show power supply detail, Expected Output: Power Supply health and redundancy status, Description: This command offers insight into the Power Supply's health and redundancy status, aiding in the diagnosis of potential power-related issues.
```

```
Command: show linecard detail, Expected Output: Line Card health and status, Description: This command provides critical information about the Line Card's health and status, assisting in the diagnosis of potential Line Card-related issues.
```

## Resolution Steps

Step 1: Step: Replace faulty Line Card, Command: replace linecard slot , Description: Replace the Line Card with a new one, ensuring compatibility with the existing system configuration.

Step 2: Step: Replace faulty Power Supply, Command: replace power-supply , Description: Replace the faulty Power Supply with a new one, ensuring redundancy and system stability.

Step 3: Step: Clear Line Card and Power Supply alarms, Command: clear system internal alarm , Description: Clear the alarms associated with the Line Card and Power Supply to prevent system instability and potential data loss.

## Post-Check Verification

```
Command: show system internal memory detail, Expected Output: No parity errors or Line Card crashes, Description: Verify that the system memory is stable, with no parity errors or Line Card crashes.

Command: show power supply detail, Expected Output: Power Supply health and redundancy status, Description: Verify that the Power Supply is healthy and redundant, ensuring system stability.

Command: show linecard detail, Expected Output: Line Card health and status, Description: Verify that the Line Card is healthy and functioning correctly.
```

## Roll-back Procedures

- Step: Restore original Line Card or Power Supply, Command: replace linecard slot , Description: Restore the original Line Card or Power Supply to prevent system instability and potential data loss.

- Step: Clear Line Card and Power Supply alarms, Command: clear system internal alarm , Description: Clear the alarms associated with the Line Card and Power Supply to prevent system instability and potential data loss.